

Stat 3113 Exam 2- Practice Problems Set 1

WARNING: THIS REVIEW IS NOT MEANT TO BE REFLECTIVE OF THE ACTUAL EXAM. THIS IS SIMPLY A SET OF SOME EXTRA PROBLEMS ON THE EXAM MATERIAL. IT IS NOT EXHAUSTIVE OF ALL TYPES OF MATERIAL OR QUESTIONS. YOU SHOULD USE THIS AS ONE STUDY TOOL, NOT YOUR ONLY STUDY TOOL!

During the test, no Excel formulas are allowed in your responses!

1. A multiple-choice test contains 10 questions (each question is unrelated), each with five choices (A, B, C, D, E). Suppose Aaron is unprepared for this test and decides to guess at each question. Let X = the number of questions answered correctly out of 10.
 - a. Is this a binomial distribution setting? Justify your answer.
 - b. What is the probability that exactly 5 questions are answered correctly?
2. Traffic engineers install 10 streetlights with new bulbs. The probability that a bulb fails within 50,000 hours of operation is 0.25. Assume that each of the bulbs fails independently.
 - a. What is the probability that fewer than two of the original bulbs fail within 50,000 hours of operation?
 - b. What is the probability that more than four of the original bulbs will need replacing within 50,000 hours?
 - c. Find the expected value. Interpret this quantity.
3. Let X denote the number of calls received per day by the MST Campus Police. Suppose that X has a Poisson distribution with mean 6.
 - a. What is the probability that, on a given day, there are at most 2 calls?
 - b. What is the probability that, in two days, 5 calls are received?
4. An automobile manufacturer is concerned about a fault in a braking mechanism of a particular model. The fault can, on rare occasions, cause a catastrophe at high speed. Suppose that the distribution of Y , the number of cars per year that will experience the catastrophe, is Poisson with mean 2.2. Find the probability that:
 - a. no cars will experience the catastrophe.
 - b. 3 or more cars will experience the catastrophe.
5. A discrete random variable X has pmf $f(x) = \begin{cases} \frac{1}{12}(6 - 2x), & x = 0, 1, 2 \\ 0, & \text{otherwise} \end{cases}$
 - a. Show that $f(x)$ is a valid pmf
 - b. Compute $E(X)$

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6. Customers arrive at a travel agency at a mean rate of 11 per hour. If the number of arrivals per hour has a Poisson distribution,
- Compute the probability that more than 3 customers will arrive each hour.
 - Find the probability that 5 customers will arrive in a one-hour period.
7. The length, X , of satisfactory service (years) provided by a certain model of laptop computer is a random variable having the probability density $f(x) = \begin{cases} \frac{1}{4.5} e^{-x/4.5}, & x \geq 0 \\ 0, & x < 0 \end{cases}$.
- Find the CDF of X .
 - Find the probability that one of these laptops will provide satisfactory service for at most 2.5 years.
 - Find the probability that one of these laptops will provide satisfactory service for at least 6.75 years.
 - What is the average length of satisfactory service?
8. Find the value of c that makes the following valid probability density functions.
- $f(x) = cx^2, 0 < x < 4$
 - $f(x) = c(1 + 2x), 0 < x < 2$
 - $f(x) = ce^{-x}, x > 0$
9. The CDF of a continuous random variable is a step function. True or False
10. Which one of the following would be a correct interpretation if you have a z-score of +2.0 on an exam?
- It means that you missed two questions on the exam.
 - It means that you got twice as many questions correct as the average student.
 - It means that your grade was 2 points higher than the mean grade on this exam.
 - It means that your grade was in the upper 2% of all grades on this exam.
 - It means that your grade is 2 standard deviations above the mean for this exam.
11. If an estimator is unbiased, then the estimator is equal to the true value it is estimating.
True/False
12. Let X denote the number of bars of service on your cell phone whenever you are at an intersection with the following probabilities.

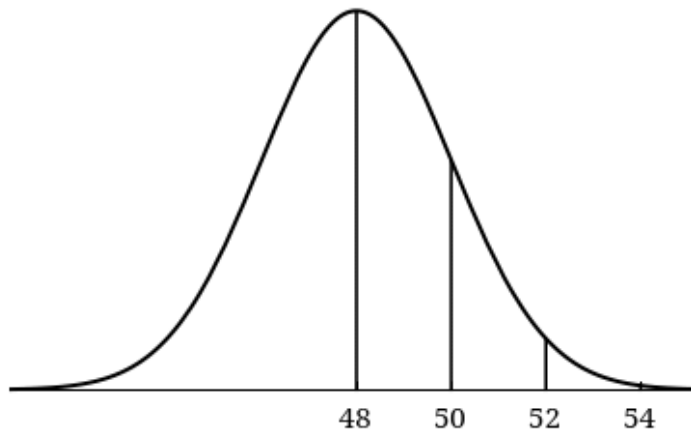
X	0	1	2	3	4	5
P(X=x)	0.1	0.15	0.25	0.25	0.15	0.1

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- a. Determine $F(x)$
 - b. What is the average number of bars of service on the cell phone?
 - c. Calculate $P(X \leq 3.5)$
13. The probability density function for the diameter of a drilled hole in mm is given as $f(x) = \begin{cases} 5e^{-5(x-3)}, & x \geq 3 \\ 0, & x < 3 \end{cases}$. Which of the following integrals represents $E(X^2)$?
 - a. $\int_3^{\infty} 5e^{-5(x-3)} dx$
 - b. $\int_3^{\infty} 5x^2 e^{-5(x-3)} dx$
 - c. $\int_3^x 5e^{-5(u-3)} du$
 - d. $\int_3^{\infty} 5xe^{-5(x-3)} dx$
14. It is often assumed that IQ scores follow a normal distribution with a mean of 100 and a standard deviation of 14. That's $X \sim N(100, 14)$. According to the empirical rule,
 - a. About 68% of scores are between 86 and 142
 - b. About 95% of scores are between 72 and 128
 - c. About 99.7% of scores are between 72 and 142
 - d. None of the above
15. Under the standard normal distribution curve, an area can be any value between $-\infty$ and ∞ .
True or False
16. The mean of the probability density function $f(x) = \begin{cases} \frac{2}{x^3}, & x > 1 \\ 0, & x \leq 1 \end{cases}$ is represented by the integral
 - a. $\int_0^1 \frac{2}{x^2} dx$
 - b. $\int_1^{\infty} \frac{2}{x^2} dx$
 - c. $\int_1^{\infty} \frac{2}{x} dx$
 - d. $\int_0^{\infty} \frac{2}{x^2} dx$
17. Let the random variable X be binomially distributed with the fixed number of trials and probability of success given as $n=7$ and $p=0.6$ respectively. Find $P(3 \leq X \leq 5)$.

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18. Suppose that the number of drivers who travel between a particular origin and destination during a designated time window, has a Poisson distribution with parameter $\lambda = 20$. What is the probability that the number of drivers will:
- Be at most 10?
 - Exceed 20?
 - Be strictly between 10 and 20?
19. Find the following quantities for the normal density curve below: $\mu, \sigma, \mu + \sigma, \mu + 2\sigma, \mu + 3\sigma$



20. What is the percentage of the area under the normal curve that lies to the right of μ ?
- 30%
 - 40%
 - 50%
 - 70%
21. What is the percentage of the area under the normal curve that lies to the left of $\mu + \sigma$?
- 16%
 - 50%
 - 68%
 - 84%
22. A normal distribution $\mu = 30$ and $\sigma = 5$. Find the z-score corresponding to $x = 25$.
- 2
 - 1
 - 0

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- d. 1
23. A normal distribution $\mu = 30$ and $\sigma = 5$. Find the raw score corresponding to $z = -2$.
- a. 20
 - b. 25
 - c. 30
 - d. 45
24. Suppose 30% of the area under the normal curve lies to the right of z . Then z is positive.
- a. True
 - b. False
25. Suppose that a normal density curve is centered at 30000 and has a standard deviation of 1000. Then the probability that a value selected at random from this distribution is greater than 30000 is 0.5.
- a. True
 - b. False
26. A random sample of size $n = 64$ is selected from a population with mean $\mu = 30$ and standard deviation $\sigma = 5$. Then the shape of the sampling distribution of the sample mean is approximately
- a. Approximately normal
 - b. Exactly normal
 - c. Approximately uniform
 - d. Exactly right skewed
 - e. Unknown
27. A random sample of size $n = 15$ is selected from a population with mean $\mu = 30$ and standard deviation $\sigma = 5$. Then the shape of the sampling distribution of the sample mean is approximately normal.
- a. True
 - b. False
28. If $f(x) = \frac{1}{4}x$; $(1 \leq x \leq 3)$ defines a valid probability density function, then the total area under the graph of the function if it is non-negative is known to be 1.
- a. True
 - b. False

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29. The total number of goals scored in a soccer game is a _____ random variable.

Discrete/Continuous?

30. Find the missing value that makes the table below a valid probability mass function.

X	0	1	2	3	4
P(x)	k	0.15	0.15	5k	0.1

31. Consider the following cumulative distribution function below. Find the $P(X \leq 3.5)$,

$P(3 \leq X \leq 6)$ and $P(X \geq 4)$.

$$F(x) = \begin{cases} 0 & x < 1 \\ 0.40 & 1 \leq x < 3 \\ 0.53 & 3 \leq x < 4 \\ 0.55 & 4 \leq x < 6 \\ 0.87 & 6 \leq x < 12 \\ 1 & x \geq 12 \end{cases}$$